



NATIONAL HARMONISED PROGRESSION SHEET FOR COMPUTER SCIENCE FORM 1

SCHOOL YEAR.....

SCHOOL.....

WEEKLY WORKLOAD: 2 periods

TEACHER.....

GRADE.....

TEL.....

Term	Week	Module	Category of action	Competency statement	Lesson title	Objectives	Resources	Nature of lesson		
								Th	Prac	Dig
FIRST TERM	1	Computing environment 1	Understanding Computer evolution	Given a situation with factors related to the evolution of computers, learners explain computer evolution and classify computers based on technology used. The explanation should bring out the inventor, date, and name of device.	Diagnostic evaluation Lesson 1: Introduction to Computer Science	- Define Computer Science, IT, ICT - Identify subareas of the field of Computer Science (Algorithms & Data Structures. Programming Languages, Architecture, Operating Systems and Networks, Software Engineering, Databases & Information Retrieval, Artificial Intelligence & Robotics...)				
					Lesson 2: History of computers	- Define a computer and trace its history by identifying major computing devices, their dates, and inventors. - Explain the concept of computer generations and list the different generations - Identify the main technology used in each generation.				
	2				Lesson 3: Classification of Computers	Classify and differentiate computers into super computer, mainframe computers, microcomputers, mini computers. Based on size, number of users, processing power and main purpose.				
					Lesson 4: Introduction to Artificial Intelligence	- Define AI - Explain the history and evolution of AI - Identify and categorise the different types of AI in real world				
	3				Lesson 5: Applications and Ethics of Artificial Intelligence	- Identify common applications of AI - Explain the uses of AI in different fields - State the ethical issues related to AI				

		Computational thinking and coding 1	Introducing analytical and logical thinking	Given a problem, learners develop strategies to solve problems logically and adapt common solutions to similar problems	Lesson 6: The use of appropriate Prompts to generate AI responses	- Examine prompt results - Discuss prompt results - Write appropriate prompts to solve problems within a given context.				
	4				Lesson 7: Integration activity					
					Lesson 8: Decomposition	- Describe computational thinking. - Apply decomposition to break a problem down into smaller parts.				
	5				Evaluation					
					Lesson 9: Pattern recognition	Apply pattern recognition.				
	6				Remediation					
					Lesson 10: Abstraction	- Apply abstraction. - Create a simple model.				
	7				Lesson 11: Introduction to algorithms	- Define an algorithm. - Use a flowchart to describe the solution to a problem.				
					Lesson 12: Integration activity					
	8	Computing environment 1	Discovering the computing environment	Presented with a computing environment, learners identify and state the roles of the main components of the environment.	Lesson 13: Components of the computing environment and role	- Identify the main components of a computing environment - Give the role of each component of a computing environment - Describe the relationship between components of a computing environment				
					Lesson 14: Users of the computing environment and notions on data and information	- List different types of users (technical and non-technical) in a computing environment and state the role of a given computer user. - List the different forms of data - Differentiate between data and information				
	9				Lesson 15: Integration activity					
			Describing the computer laboratory	Provided with a situation with factors related to setting up a computer laboratory, learners identify and enumerate the needs and rules of a school computer laboratory.	Evaluation					
	10				Remediation					
					Lesson 16: Computer laboratory equipment and devices	- Identify key elements of a computer laboratory. - State the role of each key element of a computer laboratory.				
					Lesson 17: Computer laboratory designs	- Identify the steps for a computer laboratory set up. - Identify the different layouts of a computer laboratory. - State advantages and disadvantages of different layouts.				
	11				Lesson 18: Behaviours to adopt in a computer laboratory	- Identify good behaviour to adopt when working in a computer laboratory. - State precautions to take when using a computer laboratory.				

SECOND TERM		Exploring the computer	Presented with a variety of computers and their components, learners correctly identify the types of computers involved and their respective components and effectively use the main input devices.		• State the importance of rules in a computer laboratory.						
	12			Lesson 19: Integration activity							
	13			Lesson 20: Types of personal computer	• State the different types of personal computers. • List the advantages and disadvantages of each type of personal computer.						
				Lesson 21: Components of a computer	• List the main components of a computer. • State the role of the main components of a computer. • Outline for each computer component, the different types.						
				Lesson 22: Keyboard, mouse and touch screen manipulations	• Identify actions that can be performed using a keyboard, mouse, and touchscreen • Manipulate a keyboard, mouse and touch screen						
		14	Lesson 23: Integration activity								
	15	Exploring areas of computer application	Presented with domains of life and specific computing technologies, learners describe the use of computers in these domains. The description should include an example of technologies used in the domain.	Evaluation							
				Remediation							
				Lesson 24: Application of computers	• List common areas where computers can be used. • Describe how computers can be used in different areas or domains. • List common areas where robots can be used. • Describe how robots can be used in different areas or domains.						
				Lesson 25: Application of embedded systems and IOT	• List common areas where embedded systems and IOT can be used. • Describe how embedded systems and IOT can be used in different areas or domains.						
				Lesson 26: Notions on Generative AI	• Define artificial intelligence (AI), generative AI and prompt, • Describe how to get generative AI to do exactly what you want.						
				Lesson 27: AI assisted creation	• Explain how AI works. • Evaluate how AI systems generate outputs and why they can be wrong.						
				Lesson 28: Integration activity							
				16	Digital Citizenship 1	Introducing Ethics in computing	Given a computing environment, learners adopt responsible behaviours in such an environment.	Lesson 29: Introducing Ethics	• Identify situations of unethical behaviour in life. • Demonstrate ethical behaviour.		
	Lesson 30: Computer Ethics	• Define computer ethics. • State the importance of computer ethics. • List three major issues of concern in the study of computer ethics.									
	Lesson 31: Ethical ways of using technology	• Discuss ethical ways of using technology. • Give examples of scenarios highlighting ethical issues.									
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		Computational Thinking and Coding 1	Introducing programming tools	Given a situation that requires writing a program, learners create scenarios or programs using a block programming tool.		• List common computer crimes.				
	20				Lesson 32: Integration activity					
					Evaluation					
					Remediation					
					Lesson 33: Block programming environment	• Give examples of block programming environments (e.g. Blockly, Scratch,). • Outline features of a block programming environment.				
	21	Lesson 34: Pseudocodes and Block programming	• Write algorithms using blocks.							
				Lesson 35: Write programs in a block programming environment	• Create scenarios in a block programming environment.					
	22	Lesson 36: Integration activity								
	23	Computing environment 1	Describing hardware	Given peripheral devices and a system unit, learners are required to set it up correctly and power on the computer.	Lesson 37: Peripheral devices	• Define a peripheral device • Identify and give the role of common peripheral devices (printer, speaker, microphone, scanner, projector)				
					Lesson 38: Computer ports	• Define a computer port • Identify common computer ports • Match a computer port to a connector				
					Lesson 39: The boot process	• Identify the power button • Boot a computer • Explain the boot process				
			24	Describing software	Given a set of tasks, learners identify appropriate application software needed to carry out each task.	Lesson 40: System software	• Define software, system software, application software. • Give the difference between system software and application software. • State and give examples of system software.			
Lesson 41: Application software						• State and give examples of application software. • State examples of tasks that can be carried out by a given application software.				
THIRD TERM	25	Computing environment 1	Working with GUI Operating system	Provided with a situation where files are being created or transferred into a computer, learners use basic features of an operating system to organise files logically.	Lesson 42: Integration activity					
	Evaluation									
	Remediation									
	Lesson 43: Features of a GUI				• Identify features of a GUI operating system. • Access key features of a GUI operating system.					
	26	Lesson 44: Operations on files and folders	• Give the difference between a file and a folder • State operations that can be performed on files and folders and manipulate files and folders in a GUI OS. • Arrange items in an operating system window with respect to a specific option in a pop-up menu.							
				Lesson 45: Integration activity						
	27	Using	Given a situation that	Lesson 46: Features of a word	• Define a word processor and give examples.					

		29	application software	requires the use of software to perform a task, learners create expected content using the appropriate software.	processor	- Describe key features of a word processor. - Perform simple editing and formatting of text using a word processor				
	Lesson 47: Features of a spreadsheet				- Define a spreadsheet and give examples - Describe key features of a spreadsheet - Perform basic operations using spreadsheets					
	Lesson 48: Features of a graphic software				- Define graphic software and give examples. - Describe key features of graphic software. - Perform basic operations using graphic software.					
	Lesson 49: Integration activity									
	30	31	Discovering the internet	Given a situation with issues related to searching information from the internet, learners use a web browser and hyperlinks to navigate between web pages and find information	Lesson 50: Introduction to the internet	- Define the internet and outline major landmarks in the history of the internet. - State the basic components needed to use the internet. - Explain the concepts of browser, web page, hyperlinks, URL.				
	Lesson 51: Web browsers and search engines				- View a specific website and navigate between pages and sites using hyperlinks. - Define and give examples of search engines. - Find information on the web using a search engine.					
	Lesson 52: Integration activity									
	32	Digital Citizenship 1	Computers and the workplace	Given a computing environment, learners adopt appropriate safety measures in the environment	Lesson 53: Common dangers in a workplace: Electricity shocks, fire, flood.	- Identify causes of electricity shock, fire, and flood in a computing environment. - Outline measures to prevent electricity shock, fire, and flood in a computing environment.				
	Lesson 54: Computer related health issues				- Explain the concept of repetitive strain injury (RSI). - Identify causes of repetitive strain injury. - State methods of preventing RSI.					
	Lesson 55: Integration activity									
	Evaluation									
	Remediation									
	Revision									
	36	END OF PROGRAM								